Pair-Conditioned Verification of Literature-Mined Biotic Interactions

Anonymous ACL submission

Abstract

Biotic interaction databases (who eats, infects or pollinates whom) are populated in part by text mining. Candidates are proposed when a passage names two taxa near an interaction term, and the obvious filter decides whether it describes an interaction. A passage naming several taxa may support only some of its pairs, yet this filter gives them all the same label. In our benchmark, passages naming three or more taxa hold 62% of the interacting pairs. Keeping a 110M-parameter encoder and its LLM-labelled training data fixed, we vary the input: the passage alone, with the candidate pair, or with the pair and its interaction term. On 437 expert-graded candidates, adding the pair raises the area under the precision-recall curve (AUPRC) from 0.867 to 0.942 (the two names alone reach 0.808), and the interaction term adds no significant further gain. The gain is largest on passages naming three or more taxa. It replicates on the public BioRED corpus and in zero-shot Qwen models from 1.7B to 122B parameters, which rank candidates better when asked about the pair at every size. Asked only about the passage, even the 32B model that labelled our data ranks below the 110M encoder given the pair.

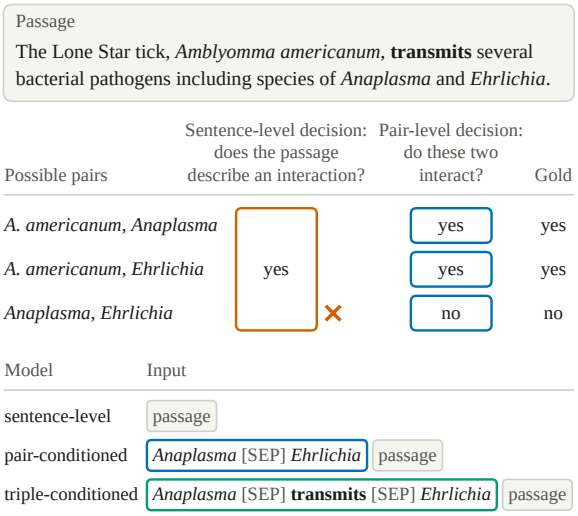


Figure 1: One passage, several candidate pairs. Retrieval proposes a candidate when a passage names two organisms near an interaction term (bold). The sentence-level decision is made once for the whole passage, so even a perfect sentence-level classifier accepts the third pair, two bacterial genera merely listed together. The third pair is a real benchmark item (TEST299, PMID 26751816, graded negative) that the sentence-level model accepts and both conditioned models reject at their block-held-out thresholds. The first two pairs are illustrative. Bottom: model inputs.

1 Introduction

Databases such as GloBI (Poelen et al., 2014) record which taxa eat, infect, parasitise or pollinate which, and, like many biomedical databases (Hirschman et al., 2012; Lee et al., 2018), they are populated in part by text mining (Dimitrova et al., 2020). A retrieval system proposes candidate triples (s_1, r, s_2), two organisms and an interaction term, with the passage that allegedly supports them, and something must decide which candidates a curator should review. A natural choice is a sentence-level classifier that reads the passage and decides whether it describes a biotic interaction.

This classifier makes one decision per passage, whereas co-occurrence retrieval proposes candidates pair by pair: a passage naming two taxa near an interaction term yields a candidate whether or not it asserts that interaction between those two. Every candidate already contains the wording a sentence-level classifier responds to: in the retrieval pool we study (§3.1), all 175,588 candidates carry an interaction term, and a lexical scan for Relation Ontology interaction terms fires on 99.83% of their passages. The decision that matters is narrower: does the passage assert an interaction between these two taxa? A passage that names several organisms can support one pair and not another, and a classifier that sees only the passage cannot tell which pair

the candidate names: even a perfect one accepts the unsupported pair (Figure 1).

Ruch et al. (2024), who built the retrieval system, graded 100 random BiotXplorer triples, found that the system identified the interacting taxa with a precision of 31%, and named among the main causes of error "instances where passages listed multiple species, which can be automatically filtered out." On our benchmark (TaxaPairs, §3.3), passages naming three or more taxa hold 156 of the 251 interacting pairs (62%), so filtering them out would discard much of what the pipeline exists to find. We keep those passages and judge each candidate by its pair.

To test this, we train the same 110M-parameter encoder three times on the same corpus (labelled by a 32B language model), reading the passage alone (*sentence-level*), the passage with the candidate pair (*pair-conditioned*), or the passage with the whole candidate triple (*triple-conditioned*), and compare them on 437 expert-graded candidates. Everything except the input is held fixed. Telling the model which pair it is judging raises the area under the precision-recall curve (AUPRC, which summarises precision over all recall levels; a random ranking scores the share of positives) from 0.867 to 0.942 and F1 from 0.790 to 0.882 (McNemar $p = 3.8 \times 10^{-7}$). At equal review workload its F1 is higher at every share of candidates accepted between 4% and 89% (Figure 2b), and adding the relation term gives no significant further gain (§4.1). On BioRED (Luo et al., 2022), a public biomedical corpus, the same comparison raises AUPRC from 0.738 to 0.843 and entity-pair F1 from 59.5 to 69.6, with the gain concentrated on sentences that name more than two entities. Zero-shot Qwen models from 1.7B to 122B parameters show the same gap, and the verifier's recommended variant (§5) runs at 32 candidates per second on a CPU.

Contributions.

- A controlled comparison of sentence-level, pair-conditioned and triple-conditioned verification on one encoder and one corpus, replicated on a public benchmark and in zero-shot Qwen language models from 1.7B to 122B parameters, with the gain concentrated on passages that name more than two entities, and a control given only the two names, which stays well below the pair-conditioned model (§4).
- An account of what giving the model the pair contributes, namely disambiguation between the organisms a passage names and a check on candidates whose arguments are not organisms, and a bound showing that checking a triple's components separately cannot reach the largest error class (§4.3, §4.6).

We also give operating curves for choosing a threshold; a practical configuration with candidate rules, a four-way direction label, CPU throughput and escalation to a zero-shot large language model (LLM) (§5; Appendices L–N); and six negative results (Appendix P).[1]

2 Related Work

Partial-input baselines. That a classifier can score well by exploiting how its data were assembled is well established in NLP. Partial-input baselines recover much of the label from the hypothesis alone in natural language inference (Gururangan et al., 2018; Poliak et al., 2018) and from the passage alone in reading comprehension (Kaushik and Lipton, 2018), and such shortcuts hold in distribution while failing under deployment (Geirhos et al., 2020; Du et al., 2024). The closest task-level precedent is fact verification (Thorne et al., 2018), where a claim-only classifier performs competitively with evidence-aware models because of how the corpus was built (Schuster et al., 2019). The sentence-level filter of §1 is also a partial-input model. Here, though, the missing input is the candidate pair the pipeline itself proposed; in those and other studies (Wiegand et al., 2019; Parmar et al., 2023), the shortcut comes from how the data were collected. Partial-input results need careful interpretation (Feng et al., 2019), and our claims are limited to the models and corpora we study.

Entity pairs in relation extraction. Distant supervision assumes that a sentence containing two related entities expresses the relation (Mintz et al., 2009), and its standard relaxation exists because it does not (Riedel et al., 2010). In supervised relation classification, as in TACRED (Zhang et al., 2017), an instance is the triple (sentence, e_1, e_2) rather than the sentence, and how the pair is injected into the encoder changes F1 by several points (Baldini Soares et al., 2019; Wu and He, 2019; Zhou and Chen, 2022). Rosenman et al. (2020) already name our failure as an "event heuristic" that

[1] Code: https://anonymous.4open.science/r/biotic-interaction-classifier-anon-EC27.

classifies on (sentence, relation) while ignoring the arguments. Our contribution is a controlled measurement of its effect in a verification setting built by co-occurrence retrieval, replicated on BioRED and in zero-shot Qwen models, and a bound on component-wise remedies. The entity-type shortcut usually invoked for such errors (Peng et al., 2020; Mtumbuka and Schockaert, 2024) is largely unavailable on TaxaPairs, where both slots hold taxon surface forms, but the names alone predict part of the label (§4.5). Error attribution across pipeline stages is standard and easy to get wrong (Taillé et al., 2020; Ai and Kavuluru, 2023).

Conditioning on the candidate. Casting a relation as a question about a named entity (Levy et al., 2017), or as an entailment hypothesis verbalising the relation between two of them (Obamuyide and Vlachos, 2018; Sainz et al., 2021), is the reformulation our input format follows, and the resulting decision (does this passage support this specific claim?) has the shape of scientific claim verification (Wadden et al., 2020), and of validating a knowledge-graph triple against text (Lehmann et al., 2012), except that those systems retrieve evidence for a given claim or triple, while here retrieval fixes the passage. Distilling a large model's judgements into a small student is standard (West et al., 2022; Gekhman et al., 2023). Selective classification abstains when a confidence score falls below a threshold (Geifman and El-Yaniv, 2017), which is how extraction is made useful to human abstractors (Swaminathan et al., 2024).

Text mining for interaction databases. Judging an extracted fact before it reaches a curator is an old idea (Rodriguez-Esteban et al., 2006), as is evaluating such tools by whether they help the curator (Hirschman et al., 2012; Britan et al., 2018; Lee et al., 2018). Cuzick et al. (2023), independently, take the interacting pair rather than the passage as the annotation unit, the unit our pair-binding error class (§4.6) isolates. Gold corpora for biodiversity text annotate entities and the relations between them, such as microorganisms with their habitats and phenotypes (Bossy et al., 2019) or organisms with environments, processes and qualities (Abdelmageed et al., 2022). They serve to train and score extraction from text, whereas our task is to verify candidates that retrieval has already proposed. Farrell et al. (2024) review text mining for ecology and evolution. Large language models mine ecological text (D'Souza et al., 2025) and extract interactions at scale (Keck et al., 2025; Zou et al., 2026). Zou et al. (2026), working from citizen-science comments, standardise names and correct categories by manual inspection, with no automatic step that checks each extraction against its text. We are not aware of published work that verifies literature-mined biotic-interaction triples against their source passage before ingestion.

3 Method

3.1 Candidates and task definition

BiotXplorer (Ruch et al., 2024) proposes candidate biotic-interaction triples from the literature indexed by SIB Literature Services (Gobeill et al., 2020). For each passage it matches taxon surface forms against a taxonomy and interaction surface forms against a vocabulary derived from the Relation Ontology (Smith et al., 2005), and emits candidate (s_1, r, s_2) triples with the passage, the matched surface strings and their canonical forms. We use it only as a source of candidates: no component of it is a baseline in this paper. The pool we study has 175,588 candidate rows over 165,325 distinct passages and 115,404 distinct triple keys. Canonical strings often differ from what the passage says, so we give the model the surface form. The decision to make is whether the passage asserts an interaction between the candidate's two organisms, in either direction (§3.5).

3.2 Training data and models

The teacher. Labels come from Qwen3-32B (Qwen Team, 2025), a locally hosted open-weight model, prompted zero-shot at temperature 0 (prompts in Appendix A) and asked, per candidate, whether the passage supports this triple. Its labels therefore vary across candidates that share a passage. Against 150 expert-graded items its mislabel rate is $22/150 = 14.7\%$ (Clopper–Pearson $[9.4\%, 21.4\%]$; 10 false positives, 12 false negatives).

The corpus. 48,338 teacher-labelled rows, 0.343 positive, drawn from the same retrieval system as the benchmark (disjoint passages): a 20,000-row base sample, 16,546 *contrast* rows, in which one passage and its pair recur with different relation terms (on 2,017 passages with labels that disagree), 6,201 actively sampled rows, and 5,591 rows with the candidate's argument order reversed. The base sample excludes the pairs of TEST299 and BIOTX100 (Appendix Q).

Models and inputs. All three are cross-encoders on BiomedBERT-base (formerly PubMedBERT; 110M parameters; Gu et al., 2021), distilled from the teacher's labels (Buciluǎ et al., 2006; Hinton et al., 2015) with the same recipe (Appendix O), keeping the checkpoint with the best AUPRC on a 10% development split grouped by taxon pair so that no pair (ignoring case and order) spans training and development. Three seeds each; one run takes about four minutes on one A100. They differ only in their input. The **sentence-level** model reads the passage alone. It is the simplest formulation, and on rows that share a passage it is shown the same input with different labels, which it cannot fit. The **pair-conditioned** model reads the two candidate taxa, s_1 `[SEP]` s_2, as segment A and the passage as segment B (Nogueira and Cho, 2019). The **triple-conditioned** model reads the whole candidate, s_1 `[SEP]` r `[SEP]` s_2. Comparing the three separates the pair's effect from the relation term's.

3.3 An expert-graded benchmark

Our benchmark, TaxaPairs, has 437 rows, 251 positive, in three blocks of different provenance. We report every block as well as the pooled figures, since pooling alone hides their disagreement (§4.3, Appendix C). TaxaPairs was built and graded by three of the authors, trained biomedical curators. BIOTX100 consists of retrieved candidates, each with one passage chosen by retrieval, expert-graded on four axes: whether each taxon surface form was correctly resolved, whether the relation was correct, and whether the whole triple is supported by its passage. Half were drawn from pairs already recorded in GloBI and half at random, and after removing three training duplicates it contributes 97 rows: 50 GloBI-matched with 43 positive, 47 random with 33 positive. REJECT50 consists of candidates an earlier filter rejected, expert-graded for the ones it should have kept: 41 clean rows, 11 positive. We use it to measure recall on candidates the earlier filter discarded (Appendix I). TEST299 is 299 expert-graded pair-level items, 0.548 positive, whose passages most often name several taxa.

The blocks do not carry the same gold: BIOTX100 and REJECT50 are also graded at the level of the whole triple, while TEST299 was expert-graded at the pair level only and its stored relation term was attached by retrieval. We therefore evaluate everywhere at the level all three share ("do these two taxa interact in this passage?"), and use the triple-level grades only to diagnose (§4.6). This can understate any benefit of triple conditioning, since a candidate with the right pair and the wrong relation counts as correct.

One caveat applies: we scored roughly a hundred model configurations on this benchmark during development, so the recipes compared here were chosen with it in view. We expect the controlled comparison of §4 to be the least affected.

3.4 Replication setting: BioRED

To test the claim on public data, we rebuild the same setting on BioRED (Luo et al., 2022), a corpus of PubMed abstracts with expert-annotated entities and relations. Candidates are generated by co-occurrence: every pair of distinct annotated concepts co-mentioned in one sentence, restricted to the eight entity-type pairs that carry relations in training, with the sentence as passage. A candidate is positive when BioRED relates the two concepts anywhere in the abstract, a document-level label whose noise every model shares. We follow the protocol of Lai et al. (2025): models train on the 500 abstracts of BioRED's training and development splits (15,640 candidates), select their epoch and threshold on its 100 test abstracts, and are scored on the 400 abstracts of the BioCreative VIII (BC8) test set (Islamaj et al., 2024). BioRED has no retrieved relation term, so only the sentence-level and pair-conditioned formulations are compared, trained with the recipe of §3.2.

3.5 Evaluation protocol

The threshold policy is fixed in advance. A model's reported operating point is either **pre-specified**, chosen on its own training-development split and never on the benchmark, or **block-held-out**: fitted on two of the three source blocks and applied to the third, so that every row is scored by a threshold that never saw it and every row is used. Comparisons between models use the same policy for both. Intervals are Clopper–Pearson for proportions and bootstrap for AUPRC. McNemar tests (McNemar, 1947) carry a continuity correction. With three seeds we report either the per-seed mean and standard deviation or a three-checkpoint ensemble, which averages the checkpoints' probabilities. The benchmark ships an `in_train` flag on 9 rows. A 5-gram Jaccard scan against all 48,338 training rows finds 3 more at Jaccard 1.000, all in BIOTX100. All 12 are excluded throughout.

The gold is pair-level and therefore order-free,

	TaxaPairs (437)		BioRED (16,606)		
Model	AUPRC	F1	AUPRC	F1	EP F1
Accept all	0.574	0.730	0.463	0.633	–
Pair-only	$0.808_{\pm.010}$	0.787	$0.710_{\pm.011}$	0.718	59.7
Sentence-level	$0.867_{\pm.006}$	0.790	$0.738_{\pm.004}$	0.718	59.5
Pair-conditioned	$0.942_{\pm.007}$	**0.882**†	**0.843**$_{\pm.004}$	**0.817**†	**69.6**
Triple-conditioned	**0.945**$_{\pm.005}$	0.867†	–	–	–

Table 1: Within each benchmark, all models share one encoder and one training corpus. Only the input differs. AUPRC: mean ± s.d. over three seeds. F1: three-checkpoint ensemble at block-held-out thresholds (TaxaPairs) or the pre-specified threshold (BioRED). EP F1: BioRED's document-level entity-pair F1 (%), per-seed mean. Accept all gives the base rate. Pair-only: the candidate pair without the passage (§4.5). †Pair- or triple-conditioned model against the sentence-level model, McNemar $p < 10^{-4}$; pair- against triple-conditioned, $p = 0.26$. BioRED has no relation term, hence no triple-conditioned model. Per-seed values (TaxaPairs): Appendices B and Q.

so a trained model that reads the candidate in order is scored as the higher of its two scores over the two argument orders, a score for whether the pair interacts in at least one direction. Otherwise a model would be penalised for rejecting a correctly paired candidate whose stated relation runs backwards, which the gold does not count as an error.

4 Results

4.1 Main results

Table 1 compares the three models on both benchmarks. On TaxaPairs, telling the model which pair a candidate is about raises per-seed AUPRC from 0.867 ± 0.006 to 0.942 ± 0.007 and, at block-held-out thresholds, F1 from 0.790 to 0.882 (precision 0.752 to 0.869, recall 0.833 to 0.896). On decision correctness the pair-conditioned model fixes 74 items and breaks 23, McNemar $p = 3.8 \times 10^{-7}$. The advantage does not depend on calibration: compared at equal review workload, the pair-conditioned model is ahead by 0.07 to 0.09 F1 whenever 50% to 70% of candidates are accepted, and the curves come within 0.01 F1 of each other only when fewer than 10% or more than 85% of candidates are accepted (Figure 2b; shares at six thresholds in Appendix C).

Adding the relation term gives no significant further gain. The triple-conditioned model's AUPRC is level with the pair-conditioned model's (0.945 against 0.942), and on decision correctness it fixes 21 items and breaks 30 ($p = 0.26$).

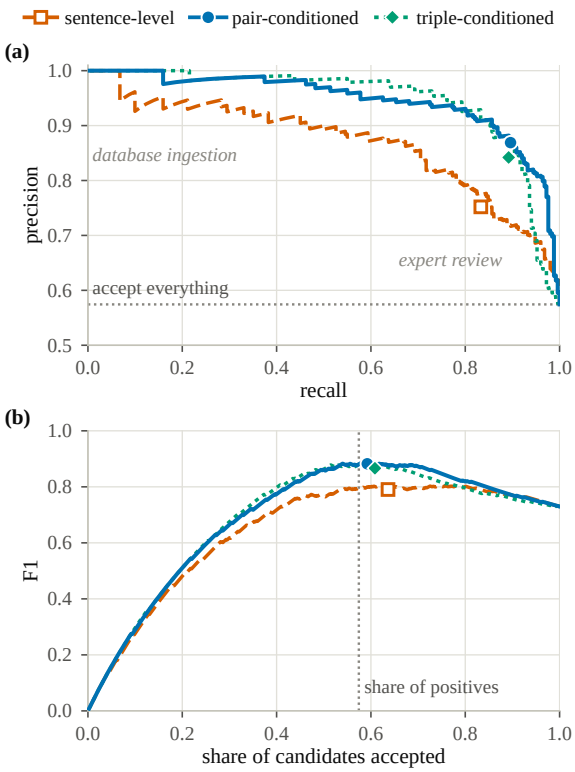


Figure 2: Operating curves on the 437 expert-graded candidates of TaxaPairs (three-checkpoint ensembles). (a) Precision against recall. Markers are the three models' operating points of Table 1. Their thresholds are fitted per block, so they can sit slightly off the pooled curve. Dotted line: accepting every candidate. (b) F1 against the share of candidates accepted, which compares the models at equal review workload whatever their calibration. The pair-conditioned model is ahead of or level with the sentence-level one at every share except 93% to 96%, where it trails by at most 0.003. Values at six thresholds: Appendix C.

4.2 Results on BioRED

The result (Table 1, right; full table in Appendix D) reproduces the one on TaxaPairs. Conditioning on the pair raises AUPRC from 0.738 to 0.843 and F1 from 0.718 to 0.817, fixing 2,642 items and breaking 1,064. Marking the two mentions in the passage, the standard relation extraction input (Zhou and Chen, 2022), adds nothing to the pair input ($p = 0.080$).

On BioRED's standard document-level metric the pair-conditioned model reaches an entity-pair F1 of 69.6, against 59.5 for the sentence-level one. Systems that read the whole abstract reach 74.1 to 75.3 (PubMedBERT, Luo et al., 2022; BioREx, Lai et al., 2023; BioREDirect; all as scored by Lai et al., 2025) and prompted GPT-4 42.9 (Islamaj et al., 2024). Part of that gap is structural: over

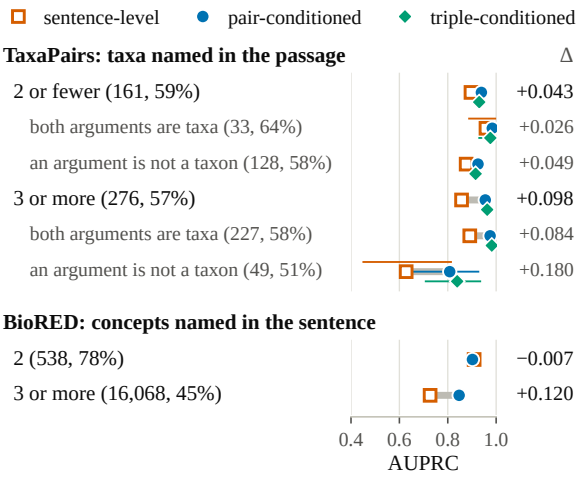


Figure 3: AUPRC by the number of entities the passage names (three-checkpoint ensembles). TaxaPairs: distinct taxa that TaxoNERD (Le Guillarme and Thuiller, 2022) finds in the passage. An argument "is not a taxon" when TaxoNERD does not recognise it there. BioRED: gold-annotated concepts in the sentence. BioRED has no relation term, so there is no triple-conditioned model. Triple-conditioned markers sit just below their row so they do not hide the pair-conditioned ones. Grey bars and the Δ column give the gain from adding the pair (pair-conditioned minus sentence-level). Parentheses give the number of candidates and the share of positives. Thin lines on the rows under 100 candidates are bootstrap 95% intervals (2,000 resamples of candidates, fixed seed; values released with the code). Exact values: Appendices C and D.

a fifth of the related pairs are never co-mentioned in one sentence, so our sentence-scope candidates cap recall at 0.776. Given the whole abstract, the same model and recipe reach 72.1 (recall 0.757, precision 0.690), two to three points below the published systems. Those also classify entity pairs, so the remaining gap reflects their document-level modelling and, for BioREx, additional training corpora, problems separate from ours.

4.3 Gain by number of taxa in the passage

Figure 3 breaks the gain down by how many taxa the passage names, which is the most direct test of our explanation. On passages that name three or more taxa, adding the pair raises AUPRC by 0.098; on passages naming two or fewer taxa, by 0.043. Splitting by argument shows where this smaller gain comes from: the candidates come from automatic matching, and 41% of the 437 have at least one "taxon" that is a string TaxoNERD does not recognise as an organism in the passage (an organ, an author's name, a vernacular term). Where both arguments are recognised taxa and the passage names no third one, the case in which the sentence-level and pair-level decisions nearly coincide, the pair input adds +0.026 on 33 items. Where an argument is not a taxon it adds +0.049, and +0.180 when the passage also names others. The pair input therefore does two things. It tells the model which pair to judge, and it lets the model check that both arguments are organisms. BioRED, whose entities are gold-annotated, isolates the first: on candidates from sentences that name only two concepts the two models are indistinguishable (0.909 against 0.902), while on those from sentences naming more, which are 97% of the candidates, conditioning adds +0.120. By source block, the pair input raises AUPRC from 0.835 to 0.944 on TEST299, whose passages name the most taxa, from 0.964 to 0.982 on BIOTX100, and from 0.551 to 0.575 on the 41-row REJECT50, too small to rank the models (Appendices C and I).

4.4 Zero-shot language models

The gap is not specific to small encoders: we put the two decisions of Figure 1 to zero-shot Qwen models as yes/no questions, the sentence question and the pair question (Figure 4; scores are $P(\text{YES})$, Appendix A). Asked the sentence question, the 32B teacher scores AUPRC 0.804, below the 110M sentence-level student; asked about the pair, 0.960. Its gap, +0.156, is twice the student's (the two questions also differ in their instructions; Limitations), and it sits in the same place: +0.179 on passages naming three or more taxa against +0.107 on the rest, where the student's is +0.098 against +0.043. The 110M pair-conditioned student, about 300 times smaller than its teacher, scores 0.14 AUPRC higher than the teacher asked the sentence question. As with the student, the relation term adds nothing at this size (0.964 against 0.960), although zero-shot models from 1.7B to 30B-A3B score higher on the triple question (Appendix E). From Qwen3-1.7B to Qwen3.5-122B-A10B (Qwen Team, 2026) the pair question leads at each size, by +0.04 to +0.20 AUPRC on TaxaPairs and +0.11 to +0.30 on BioRED, while the sentence question gains little with size (Appendix E). Against its teacher's labelling prompt (0.964) the pair-conditioned student gives up almost no F1 (0.882 against 0.885 at block-held-out thresholds) but 0.021 AUPRC, about a quarter of which soft labels recover (Appendix F).

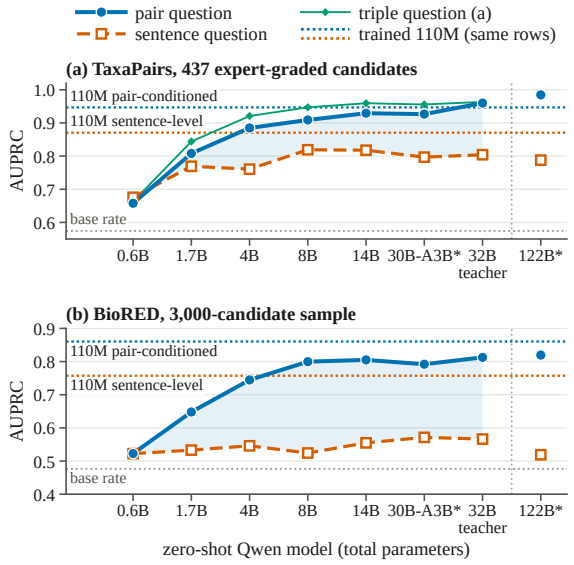


Figure 4: Zero-shot Qwen models (Qwen Team, 2025) on (a) TaxaPairs, 437 expert-graded candidates, and (b) a 3,000-candidate BioRED sample. Each panel shows answers to the sentence question (dashed, open squares) and the pair question (solid, filled circles). Dotted: the trained 110M models on the same rows (three-checkpoint ensembles); grey: base rate. The x-axis spaces model sizes evenly. From 1.7B on, the pair question leads at every size, while answers to the sentence question change little with scale. *Mixture of experts, placed by total size. The 32B model is the teacher. Qwen3.5-122B-A10B is a later release. Thin green line in (a): the triple question (TaxaPairs only), which matches or exceeds the pair question (Appendix E).

4.5 Controls: input ablation, training data, encoder, names alone

Three controls locate the effect in the pair input. Degrading the trained triple-conditioned model's segment A at inference, weights and thresholds unchanged, shows which part of segment A it uses: deleting both taxa drops the three-checkpoint ensemble's AUPRC from 0.948 to 0.691 against a base rate of 0.574, and emptying segment A gives 0.649, while replacing or shuffling the relation, pair kept, moves it far less (0.906, 0.930; Appendix G). The triple-conditioned model does not need the corpus's contrast rows: trained on the 20,000-row base sample alone, it reaches AUPRC 0.943 ± 0.004 (full corpus 0.945 ± 0.005) and block-held-out F1 0.886. The encoder is not the cause either: of seven alternative recipes screened on development data, the three that passed were scored once on test, none gains significantly on both benchmarks, and with every encoder trained in both formulations the pair input wins, by $+0.103$ to $+0.113$ AUPRC on BioRED and $+0.051$ to $+0.076$ on TaxaPairs (Appendix H).

A fourth control asks how much the two names predict on their own. A model trained with the same recipe on the pair without the passage (*pair-only*) reaches AUPRC 0.808 on TaxaPairs and 0.710 on BioRED (Table 1), well above the base rates, and 0.134 below the pair-conditioned model on TaxaPairs (bootstrap 95% interval $[0.085, 0.185]$). Only 38 of the 437 candidates have a pair that occurs in the training corpus (ignoring case and order). On the other 399, adding the pair still raises AUPRC from 0.870 to 0.951 (three-checkpoint ensembles), and on the 98 neither of whose taxa occurs in training the sentence-level, pair-conditioned and pair-only models reach 0.920, 0.980 and 0.772. Trained for up to ten epochs instead of three, the pair-only model reaches 0.804, so more epochs do not close the gap (Appendix Q).

4.6 Error analysis

The pair-conditioned model's 34 false positives at its block-held-out operating point are not homogeneous: 24 come from passages naming three or more taxa, and 16 have an argument TaxoNERD does not recognise as a taxon. Read by hand, they sort into a few mechanisms (examples in Table 10, Appendix J; the per-row table will be released with the benchmark): *pair-binding*, where both taxa and the relation are correct but the relation's argument is a third organism; *co-listing*, where the two taxa are members of one enumeration and the passage states no interaction between them; *negation*, where the passage asserts the opposite; *same organism*, where the two "taxa" are one organism named twice or an organism and its own clade; relation terms that name no biotic interaction (*migrating*); and *entity* failures, where a taxon string is a taxonomic author, a place, an organ, or an adjective inside a pathogen's name ("*avian* schistosomes", which this model already rejects).

A bound on component-wise verification. The pair-binding class limits a natural alternative architecture. Its members have both taxa correctly resolved and the correct relation. The error is a pairing made by retrieval that the passage does not state. A verifier that checks s_1, r and s_2 separately, however good each check is, cannot see it. On the 97 clean BIOTX100 candidates, where triple-level grades make the classes countable, 15 of the 32 unsupported candidates are pair-binding

failures (the rest: 9 entity and 8 relation failures), a share of 0.47, Clopper–Pearson $[0.29, 0.65]$. A component-wise verifier therefore accepts these 15 at the same rate as the 65 supported triples, so its precision is at most $65/(65+15) = 0.81$ at any recall, $[0.76, 0.87]$ across that interval. Only a model that reads the pair together with its passage can exceed this bound.

4.7 Fresh candidates

We also scored 100 candidates from outside TaxaPairs: two further batches of 50 from the same retrieval system, graded once at the pair level in the same expert curation as TEST299, 32 of them positive (a base rate of 0.32, against 0.574 on the benchmark). None of their passages occurs in the benchmark or in the training corpus. As on the benchmark, segment A uses the taxon and relation strings as they appear in the passage. All three models score below their benchmark figures, and the sentence-level model is again last: AUPRC 0.665 sentence-level, 0.787 pair-conditioned and 0.751 triple-conditioned (three-checkpoint ensembles), a lead of 0.122 for the pair-conditioned model (bootstrap 95% interval $[0.010, 0.207]$). A lower base rate lowers AUPRC by itself, but AUROC, which does not depend on it, drops too (0.805, 0.878 and 0.843, against 0.842, 0.932 and 0.919 on the benchmark), so the benchmark is likely easier than a live pool and its absolute figures optimistic.

5 Discussion and Conclusion

Partial input and model size. The remedy for a partial-input filter is to give it the missing input, here the candidate pair the pipeline proposed. Larger models do not make up for it. From 1.7B to 122B, zero-shot Qwen models answer the pair question better than the sentence question at each size, and their answers to the latter gain little with size. A 110M model given the pair ranks better than the 32B teacher asked about the passage (§4.4). The opposite partial input, the pair without its passage, also ranks below the two together (§4.5).

Passages naming only two taxa. The two decisions coincide when a passage names only the candidate's two entities and both are what they claim to be: on BioRED sentences that name two concepts the two models are indistinguishable, and on TaxaPairs the gain is smallest where both arguments are recognised taxa and no third is named (§4.3). Most of the positives in TaxaPairs come from passages that name more.

Choosing an operating point. The precision of the pair-conditioned model ranges from 0.761 at threshold $\tau = 0.01$ to 0.981 at $\tau = 0.90$ (Appendix C), so the right threshold depends on the use. Ingestion into a reference database calls for high precision, and assembling candidates for expert review for high recall. Reporting a score leaves this choice to the user (Geifman and El-Yaniv, 2017).

A practical configuration. For use in a pipeline we recommend a variant of the pair-conditioned model that adds deterministic candidate rules and a head labelling the interaction's direction in four categories (Appendices L and M). Its input and training labels also differ from those of the pair-conditioned model, so it lies outside the controlled comparison. It scores 32 candidates per second on a CPU, and it can escalate the candidates it is least sure of to a zero-shot LLM (Appendix N).

Errors made before verification. A wrong interaction term or a mis-resolved taxon arises during retrieval, before any verifier runs, and a model that reads the passage cannot see that the resolved taxon is not the one the passage names (Limitations).

Other domains. The problem arises in any pipeline that proposes candidates by co-occurrence and filters them one passage at a time. On BioRED the pair input also helps for genes, chemicals and diseases, though part of that gain may come from entity types (Appendix Q). For a new pipeline, the first quantity to measure is the share of passages that name more than the candidate pair.

Conclusion. When candidates come from co-occurrence retrieval, the model that verifies them should see the candidate pair as well as the passage. A classifier that reads only the passage judges every pair in it alike, and 62% of the interacting pairs in TaxaPairs come from passages naming three or more taxa. Giving the model the pair raises AUPRC from 0.867 to 0.942, with the largest gain on those passages. The effect replicates on BioRED and in zero-shot Qwen models from 1.7B to 122B parameters. The verifier runs on a CPU, fast enough to score every candidate in the pool.

Limitations

Benchmark size and annotation. TaxaPairs has 437 expert-graded rows in three blocks, each graded once. One of the authors, blind to the labels and model scores, re-graded 22 candidates: 11 on which two models (triple-conditioned and recommended, §5) confidently disagreed with the gold, and 11 random controls. The re-grade agreed with all 11 control labels (11/11, our only agreement figure) and changed 7 of the other 11. One more label had been changed earlier, from positive to negative, after one of the authors found that its TEST299 passage names *Acanthamoeba* and *Pseudomonas* only as agents found together in the same horses. On the labels before the re-grade, which include that change, the pair input raises AUPRC from 0.851 to 0.918. On the recall side (REJECT50, Appendix I) the denominator is 11 positives.

Rule selection. Every candidate rule was written with errors on this benchmark in view: four after reading its false positives, *co_listed* after the other seven had been evaluated on it, and three from an earlier filter's errors on two of its blocks (Appendix L). The training-data criterion limits how far the rules can be tailored to the benchmark but cannot exclude such tailoring, so the rules' effects should be read as optimistic until they are confirmed on candidates that were not consulted when the rules were written. On the fresh candidates of §4.7 the rules fire too rarely to confirm or contradict these effects (Appendix L).

Fresh candidates. On 100 fresh candidates from the same retrieval system (32 positive), all three models score below their benchmark figures (§4.7); part of the drop follows from the lower base rate, but the absolute figures on TaxaPairs are likely to be optimistic.

Pair-level evaluation. TEST299, the largest block, carries no whole-triple gold. Evaluating at the pair level everywhere means a candidate with the right pair and the wrong relation counts as correct, so the comparison cannot credit the relation term with the cases it alone could catch; measuring those would need triple-level gold on TEST299.

Document-level labels in BioRED. A candidate is positive when the abstract relates the two concepts anywhere, so a co-mention in a sentence that does not state the relation counts as positive. Both models are trained and scored on the same noisy labels, so the comparison is controlled, but the absolute figures need not carry over to labels judged per sentence. BioRED candidates also share sentences, so candidate-level McNemar tests and bootstrap intervals there are optimistic for small differences.

Prompt wording. Beyond naming the pair, the zero-shot pair question on TaxaPairs (Appendix A) asks the model to check that both strings name organisms in the passage and to reject a mere co-mention or an interaction each has with a third organism, and it omits the sentence question's examples of interactions (on BioRED its only added instruction is the co-mention clause), so part of its lead in §4.4 may come from this wording rather than from naming the pair. The trained models read no instructions and learn from the same labels; their inputs differ only by the pair and, for the triple-conditioned model, the relation term, so the controlled comparison of Table 1 is unaffected.

Teacher noise and pretraining exposure. The training labels come from a prompted teacher wrong about one time in seven, and the student inherits that boundary; training on model-generated labels can degrade encoders (Lu and Smith, 2025). The passages are published PubMed text and the teacher is a pretrained model, so we cannot exclude that it saw them, and its labels may be optimistic for genuinely new literature. Asked about the pair without the passage, the teacher already ranks the TaxaPairs candidates at AUPRC 0.871 (Appendix Q), so its labels may also reflect what it knows about the two taxa.

Upstream failures. Most REJECT50 candidates carry an incorrect interaction term or a misresolved taxon (Appendix I). Those fail before any verifier is consulted, and no change to the decision stage, ours included, addresses them.

Single retrieval system and domain. We study one retrieval system with one interaction vocabulary, on interactions between taxa in PubMed titles and abstracts. The mechanism, a pair-agnostic decision over passages that name several organisms, is not specific to this pool, but beyond BioRED's gold-entity candidates (§3.4) we have not shown that pools from other retrieval systems fail the same way.

References

Nora Abdelmageed, Felicitas Löffler, Leila Feddoul, Alsayed Algergawy, Sheeba Samuel, Jitendra Gaikwad, Anahita Kazem, and Birgitta König-Ries. 2022. BiodivNERE: Gold standard corpora for named entity recognition and relation extraction in the biodiversity domain. *Biodiversity Data Journal*, 10:e89481.

Xuguang Ai and Ramakanth Kavuluru. 2023. End-to-end models for chemical–protein interaction extraction: Better tokenization and span-based pipeline strategies. In *2023 IEEE 11th International Conference on Healthcare Informatics (ICHI)*, pages 610–618. IEEE.

Livio Baldini Soares, Nicholas FitzGerald, Jeffrey Ling, and Tom Kwiatkowski. 2019. Matching the blanks: Distributional similarity for relation learning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2895–2905, Florence, Italy. Association for Computational Linguistics.

Robert Bossy, Louise Deléger, Estelle Chaix, Mouhamadou Ba, and Claire Nédellec. 2019. Bacteria Biotope at BioNLP Open Shared Tasks 2019. In *Proceedings of the 5th Workshop on BioNLP Open Shared Tasks*, pages 121–131, Hong Kong, China. Association for Computational Linguistics.

Aurore Britan, Isabelle Cusin, Valérie Hinard, Luc Mottin, Emilie Pasche, Julien Gobeill, Valentine Rech de Laval, Anne Gleizes, Daniel Teixeira, Pierre-André Michel, Patrick Ruch, and Pascale Gaudet. 2018. Accelerating annotation of articles via automated approaches: evaluation of the neXtA5 curation-support tool by neXtProt. *Database*, 2018:bay129.

Cristian Buciluǎ, Rich Caruana, and Alexandru Niculescu-Mizil. 2006. Model compression. In *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 535–541.

Alayne Cuzick, James Seager, Valerie Wood, Martin Urban, Kim Rutherford, and Kim E. Hammond-Kosack. 2023. A framework for community curation of interspecies interactions literature. *eLife*, 12:e84658.

Mariya Dimitrova, Jorrit Poelen, Georgi Zhelezov, Teodor Georgiev, Donat Agosti, and Lyubomir Penev. 2020. Semantic publishing enables text mining of biotic interactions. *Biodiversity Information Science and Standards*, 4:e59036.

Jennifer D'Souza, Zachary Laubach, Tarek Al Mustafa, Sina Zarrieß, Robert Frühstückl, and Phyllis Illari. 2025. Mining for species, locations, habitats, and ecosystems from scientific papers in invasion biology: A large-scale exploratory study with large language models. In *Proceedings of the 1st Workshop on Ecology, Environment, and Natural Language Processing (NLP4Ecology2025)*, pages 16–23, Tallinn, Estonia. University of Tartu Library.

Mengnan Du, Fengxiang He, Na Zou, Dacheng Tao, and Xia Hu. 2024. Shortcut learning of large language models in natural language understanding. *Communications of the ACM*, 67(1):110–120.

Maxwell J. Farrell, Nicolas Le Guillarme, Liam Brierley, Bronwen Hunter, Daan Scheepens, Anna Willoughby, Andrew Yates, and Nicole Mideo. 2024. The changing landscape of text mining: a review of approaches for ecology and evolution. *Proceedings of the Royal Society B: Biological Sciences*, 291(2027):20240423.

Shi Feng, Eric Wallace, and Jordan Boyd-Graber. 2019. Misleading failures of partial-input baselines. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5533–5538, Florence, Italy. Association for Computational Linguistics.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 4878–4887.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.

Zorik Gekhman, Jonathan Herzig, Roee Aharoni, Chen Elkind, and Idan Szpektor. 2023. TrueTeacher: Learning factual consistency evaluation with large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2053–2070, Singapore. Association for Computational Linguistics.

Martin Gerner, Goran Nenadic, and Casey M. Bergman. 2010. LINNAEUS: A species name identification system for biomedical literature. *BMC Bioinformatics*, 11:85.

Julien Gobeill, Déborah Caucheteur, Pierre-André Michel, Luc Mottin, Emilie Pasche, and Patrick Ruch. 2020. SIB Literature Services: RESTful customizable search engines in biomedical literature, enriched with automatically mapped biomedical concepts. *Nucleic Acids Research*, 48(W1):W12–W16.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2021. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112, New Orleans, Louisiana. Association for Computational Linguistics.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*.

Lynette Hirschman, Gully A. P. C. Burns, Martin Krallinger, Cecilia Arighi, K. Bretonnel Cohen, Alfonso Valencia, Cathy H. Wu, Andrew Chatr-Aryamontri, Karen G. Dowell, Eva Huala, Anália Lourenço, Robert Nash, Anne-Lise Veuthey, Thomas Wiegers, and Andrew G. Winter. 2012. Text mining for the biocuration workflow. *Database*, 2012:bas020.

Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. spaCy: Industrial-strength natural language processing in Python. Zenodo.

Rezarta Islamaj, Po-Ting Lai, Chih-Hsuan Wei, Ling Luo, Tiago Almeida, Richard A. A. Jonker, Sofia I. R. Conceição, Diana F. Sousa, Cong-Phuoc Phan, Jung-Hsien Chiang, Jiru Li, Dinghao Pan, Wilailack Meesawad, Richard Tzong-Han Tsai, M. Janina Sarol, Gibong Hong, Airat Valiev, Elena Tutubalina, Shao-Man Lee, and 4 others. 2024. The overview of the BioRED (Biomedical Relation Extraction Dataset) track at BioCreative VIII. *Database*, 2024:baae069.

Divyansh Kaushik and Zachary C. Lipton. 2018. How much reading does reading comprehension require? A critical investigation of popular benchmarks. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 5010–5015, Brussels, Belgium. Association for Computational Linguistics.

François Keck, Henry Broadbent, and Florian Altermatt. 2025. Extracting massive ecological data on state and interactions of species using large language models. *bioRxiv*. Preprint, posted 27 January 2025.

Po-Ting Lai, Chih-Hsuan Wei, Ling Luo, Qingyu Chen, and Zhiyong Lu. 2023. BioREx: Improving biomedical relation extraction by leveraging heterogeneous datasets. *Journal of Biomedical Informatics*, 146:104487.

Po-Ting Lai, Chih-Hsuan Wei, Shubo Tian, Robert Leaman, and Zhiyong Lu. 2025. Enhancing biomedical relation extraction with directionality. *Bioinformatics*, 41(Supplement_1):i68–i76.

Nicolas Le Guillarme and Wilfried Thuiller. 2022. TaxoNERD: Deep neural models for the recognition of taxonomic entities in the ecological and evolutionary literature. *Methods in Ecology and Evolution*, 13(3):625–641.

Kyubum Lee, Maria Livia Famiglietti, Aoife McMahon, Chih-Hsuan Wei, Jacqueline Ann Langdon MacArthur, Sylvain Poux, Lionel Breuza, Alan Bridge, Fiona Cunningham, Ioannis Xenarios, and Zhiyong Lu. 2018. Scaling up data curation using deep learning: An application to literature triage in genomic variation resources. *PLOS Computational Biology*, 14(8):e1006390.

Jens Lehmann, Daniel Gerber, Mohamed Morsey, and Axel-Cyrille Ngonga Ngomo. 2012. DeFacto – deep fact validation. In *The Semantic Web – ISWC 2012*, volume 7649 of *Lecture Notes in Computer Science*, pages 312–327. Springer.

Omer Levy, Minjoon Seo, Eunsol Choi, and Luke Zettlemoyer. 2017. Zero-shot relation extraction via reading comprehension. In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)*, pages 333–342, Vancouver, Canada. Association for Computational Linguistics.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*.

Yucheng Lu and Kazimier Smith. 2025. Feeding LLM annotations to BERT classifiers at your own risk. *arXiv preprint arXiv:2504.15432*.

Ling Luo, Po-Ting Lai, Chih-Hsuan Wei, Cecilia N. Arighi, and Zhiyong Lu. 2022. BioRED: a rich biomedical relation extraction dataset. *Briefings in Bioinformatics*, 23(5):bbac282.

Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Mike Mintz, Steven Bills, Rion Snow, and Dan Jurafsky. 2009. Distant supervision for relation extraction without labeled data. In *Proceedings of the Joint Conference of the 47th Annual Meeting of the ACL and the 4th International Joint Conference on Natural Language Processing of the AFNLP*, pages 1003–1011, Suntec, Singapore. Association for Computational Linguistics.

Frank Martin Mtumbuka and Steven Schockaert. 2024. Entity or relation embeddings? An analysis of encoding strategies for relation extraction. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 6003–6022, Miami, Florida, USA. Association for Computational Linguistics.

Rodrigo Nogueira and Kyunghyun Cho. 2019. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*.

Abiola Obamuyide and Andreas Vlachos. 2018. Zero-shot relation classification as textual entailment. In *Proceedings of the First Workshop on Fact Extraction and VERification (FEVER)*, pages 72–78, Brussels, Belgium. Association for Computational Linguistics.

Mihir Parmar, Swaroop Mishra, Mor Geva, and Chitta Baral. 2023. Don't blame the annotator: Bias already starts in the annotation instructions. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1779–1789, Dubrovnik, Croatia. Association for Computational Linguistics.

Hao Peng, Tianyu Gao, Xu Han, Yankai Lin, Peng Li, Zhiyuan Liu, Maosong Sun, and Jie Zhou. 2020. Learning from context or names? An empirical study on neural relation extraction. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3661–3672, Online. Association for Computational Linguistics.

Jorrit H. Poelen, James D. Simons, and Chris J. Mungall. 2014. Global biotic interactions: An open infrastructure to share and analyze species-interaction datasets. *Ecological Informatics*, 24:148–159.

Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. Hypothesis only baselines in natural language inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 180–191, New Orleans, Louisiana. Association for Computational Linguistics.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Qwen Team. 2026. Qwen3.5: Towards native multimodal agents. https://qwen.ai/blog?id=qwen3.5. Model card: https://huggingface.co/Qwen/Qwen3.5-122B-A10B.

Jonathan A. Rees and Karen Cranston. 2017. Automated assembly of a reference taxonomy for phylogenetic data synthesis. *Biodiversity Data Journal*, 5:e12581.

Sebastian Riedel, Limin Yao, and Andrew McCallum. 2010. Modeling relations and their mentions without labeled text. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2010)*, volume 6323 of *Lecture Notes in Computer Science*, pages 148–163, Barcelona, Spain. Springer Berlin Heidelberg.

Raul Rodriguez-Esteban, Ivan Iossifov, and Andrey Rzhetsky. 2006. Imitating manual curation of text-mined facts in biomedicine. *PLoS Computational Biology*, 2(9):e118.

Shachar Rosenman, Alon Jacovi, and Yoav Goldberg. 2020. Exposing shallow heuristics of relation extraction models with challenge data. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3702–3710, Online. Association for Computational Linguistics.

Patrick Ruch, Emilie Pasche, Pierre-André Michel, and Déborah Caucheteur. 2024. BiotXplorer: Navigating evidence-based biotic interactions. *Biodiversity Information Science and Standards*, 8:e135453.

Oscar Sainz, Oier Lopez de Lacalle, Gorka Labaka, Ander Barrena, and Eneko Agirre. 2021. Label verbalization and entailment for effective zero and few-shot relation extraction. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1199–1212, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Tal Schuster, Darsh Shah, Yun Jie Serene Yeo, Daniel Roberto Filizzola Ortiz, Enrico Santus, and Regina Barzilay. 2019. Towards debiasing fact verification models. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3419–3425, Hong Kong, China. Association for Computational Linguistics.

Barry Smith, Werner Ceusters, Bert Klagges, Jacob Köhler, Anand Kumar, Jane Lomax, Chris Mungall, Fabian Neuhaus, Alan L. Rector, and Cornelius Rosse. 2005. Relations in biomedical ontologies. *Genome Biology*, 6(5):R46.

Akshay Swaminathan, Ivan Lopez, William Wang, Ujwal Srivastava, Edward Tran, Aarohi Bhargava-Shah, Janet Y. Wu, Alexander L. Ren, Kaitlin Caoili, Brandon Bui, Layth Alkhani, Susan Lee, Nathan Mohit, Noel Seo, Nicholas Macedo, Winson Cheng, Charles Liu, Reena Thomas, Jonathan H. Chen, and Olivier Gevaert. 2024. Selective prediction for extracting unstructured clinical data. *Journal of the American Medical Informatics Association*, 31(1):188–197.

Bruno Taillé, Vincent Guigue, Geoffrey Scoutheeten, and Patrick Gallinari. 2020. Let's stop incorrect comparisons in end-to-end relation extraction! In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3689–3701, Online. Association for Computational Linguistics.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, New Orleans, Louisiana. Association for Computational Linguistics.

David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, Online. Association for Computational Linguistics.

Peter West, Chandra Bhagavatula, Jack Hessel, Jena D. Hwang, Liwei Jiang, Ronan Le Bras, Ximing Lu, Sean Welleck, and Yejin Choi. 2022. Symbolic knowledge distillation: from general language models to commonsense models. In *Proceedings of the 2022 Conference of the North American Chapter of*

the Association for Computational Linguistics: Human Language Technologies, pages 4602–4625, Seattle, United States. Association for Computational Linguistics.

Michael Wiegand, Josef Ruppenhofer, and Thomas Kleinbauer. 2019. Detection of abusive language: the problem of biased datasets. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 602–608, Minneapolis, Minnesota. Association for Computational Linguistics.

Shanchan Wu and Yifan He. 2019. Enriching pre-trained language model with entity information for relation classification. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 2361–2364, Beijing, China. Association for Computing Machinery.

Michihiro Yasunaga, Jure Leskovec, and Percy Liang. 2022. LinkBERT: Pretraining language models with document links. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8003–8016, Dublin, Ireland. Association for Computational Linguistics.

Yuhao Zhang, Victor Zhong, Danqi Chen, Gabor Angeli, and Christopher D. Manning. 2017. Position-aware attention and supervised data improve slot filling. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 35–45, Copenhagen, Denmark. Association for Computational Linguistics.

Wenxuan Zhou and Muhao Chen. 2022. An improved baseline for sentence-level relation extraction. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 161–168, Online only. Association for Computational Linguistics.

Heng-Xing Zou, Xiaohao Yang, Thabassum H. Hajamaideen, Olivia J. Stein, Roxanne S. Beltran, Benjamin G. Freeman, Mark Lindquist, Eliot T. Miller, Summer Mengarelli, Charlotte M. Probst, Fernanda S. Valdovinos, Derek Van Berkel, Phoebe L. Zarnetske, Brian C. Weeks, and Kai Zhu. 2026. Large language models unlock the ecology of species interactions. *Proceedings of the National Academy of Sciences*, 123(41):e2602244123.

A Teacher prompts

The three questions put to the 32B teacher and the other zero-shot models (§4.4, Appendix E) are given below verbatim. They are asked at temperature 0 with thinking disabled, and the score is the probability of YES among the YES/NO alternatives of the first answer token. The triple question is the prompt that labelled the training corpus. On BioRED the sentence and pair questions name biomedical entities and relation types in place of organisms and biotic interactions, and there the pair question asks only whether the sentence states a relation between the two entities beyond a mere co-mention (their exact text is released with the code). Lines are wrapped for the page.

Sentence question.

```
You check sentences retrieved from the
scientific literature for a
biotic-interaction database.

Sentence: <passage>

Answer YES only if the sentence asserts
a direct biological interaction between
organisms (for example predation,
parasitism, infection, herbivory,
pollination, mutualism, symbiosis or
disease transmission), and the
interaction is not negated.

Reply with exactly one word: YES or NO.
```

Pair question.

```
You verify candidate biotic-interaction
pairs extracted from scientific
literature.

Sentence: <passage>

Candidate pair:
  Entity 1: "<taxon 1>"
  Entity 2: "<taxon 2>"

Answer YES only if ALL of the following
hold:
 1. Both entities are genuinely the
    organisms referred to by those
    surface strings in this sentence
    (not a gene, protein, chemical,
    author name, or a different
    species).
 2. The sentence asserts a direct
    biological interaction between these
    two organisms (not merely a
    co-mention, and not an interaction
    each has with some third organism),
    and the interaction is not negated.

Reply with exactly one word: YES or NO.
```

Triple question (the labelling prompt).

```
You verify candidate biotic-interaction
triples extracted from scientific
literature.

Sentence: <passage>

Candidate triple:
  Entity 1: "<taxon 1>"
  Relation: "<relation>"
  Entity 2: "<taxon 2>"
```

	AUPRC			F1, $\tau = 0.5$		
seed	sent.	pair	triple	sent.	pair	triple
1	.871	.946	.948	.634	.804	.859
2	.868	.946	.947	.708	.772	.864
3	.860	.935	.939	.670	.802	.852
mean	.867	.942	.945	.671	.793	.858
s.d.	.006	.007	.005	.037	.018	.006

Table 2: Per-seed figures on the 437 TaxaPairs rows, order-free scoring. The body reports F1 of three-checkpoint ensembles at block-held-out thresholds; a fixed $\tau = 0.5$ is shown here only for the spread across seeds, not to compare the models, because the three models are calibrated differently, and a single shared threshold would favour one of them arbitrarily.

	sentence				pair				triple			
τ	P	R	F1	Share	P	R	F1	Share	P	R	F1	Share
0.01	.709	.920	.801	.746	.761	.976	.855	.737	.740	.940	.828	.730
0.04	.782	.817	.799	.600	.815	.948	.877	.668	.811	.924	.864	.654
0.10	.819	.737	.776	.517	.854	.908	.880	.611	.845	.912	.877	.620
0.25	.861	.693	.768	.462	.911	.853	.881	.538	.903	.849	.875	.540
0.50	.886	.526	.660	.341	.940	.689	.795	.421	.942	.777	.852	.474
0.90	.930	.159	.272	.098	.981	.422	.591	.247	.982	.637	.773	.373

Table 3: Values behind Figure 2: the three models' operating curves on all 437 rows, three-checkpoint ensembles; Share is the fraction of the 437 candidates the model accepts at threshold τ. Rows show the shape; the reported operating points are those of Table 1.

```
Answer YES only if ALL of the following
hold:
 1. Both entities are genuinely the
    organisms referred to by those
    surface strings in this sentence
    (not a gene, protein, chemical,
    author name, or a different
    species).
 2. The sentence asserts a direct
    biological interaction between these
    two organisms (not merely a
    co-mention, and not an interaction
    each has with some third organism).
 3. The stated relation is the correct
    type of interaction between them, in
    the stated direction, and is not
    negated.

Reply with exactly one word: YES or NO.
```

B Per-seed results

Table 2 gives the per-seed AUPRC behind the TaxaPairs means of Table 1.

C Values behind Figures 2 and 3

Table 3 gives the operating curves of Figure 2 at six thresholds, and Table 4 the AUPRC values of Figure 3, with AUPRC by source block.

Candidates	n	Pos.	Sentence	Pair	Triple
passage names ≤ 2 taxa	161	0.590	0.896	**0.939**	0.930
passage names ≥ 3 taxa	276	0.565	0.857	0.955	**0.963**
both arguments are taxa, ≤ 2	33	0.636	0.958	**0.984**	0.976
both arguments are taxa, ≥ 3	227	0.577	0.891	0.975	**0.982**
an argument is not a taxon, ≤ 2	128	0.578	0.876	**0.925**	0.915
an argument is not a taxon, ≥ 3	49	0.510	0.628	0.808	**0.839**
TEST299	299	0.548	0.835	0.944	**0.951**
BIOTX100	97	0.784	0.964	**0.982**	0.973
REJECT50	41	0.268	0.551	0.575	**0.675**

Table 4: Exact values of Figure 3 (TaxaPairs rows) and AUPRC by source block; all values are AUPRC of three-checkpoint ensembles, and Pos. is the share of positives in the row, the AUPRC of a random ranking. Taxa are counted as distinct mentions found in the passage by TaxoNERD (Le Guillarme and Thuiller, 2022) (`en_ner_eco_biobert`); an argument "is not a taxon" when TaxoNERD does not recognise it in the passage. TEST299 passages name 5.1 taxa on average, BIOTX100 2.6 and REJECT50 1.7.

D BioRED in full

Table 5 gives the BioRED comparison of §4.2 in full: AUPRC by the number of concepts the sentence names (§4.3), the entity-marker variant, and the zero-shot teacher on a 3,000-candidate sample (§4.4). On sentences that name two concepts, the per-seed AUPRC ranges from 0.899 to 0.919 for the sentence-level model and from 0.892 to 0.914 for the pair-conditioned model.

E Zero-shot scaling

Table 6 gives the zero-shot comparison of Figure 4 in full, with the triple question added. Each model's score is $P(\text{YES})$ from its first answer token, and the triple question is the teacher's labelling prompt, asked in both argument orders and scored by the higher of the two, like the trained triple-conditioned model; the pair question is asked in one argument order. At block-held-out thresholds the 32B teacher's precision, recall and F1 are 0.654, 0.948 and 0.774 asked the sentence question, 0.919, 0.865 and 0.891 asked the pair question, and 0.878, 0.892 and 0.885 asked the triple question. On BioRED the teacher, asked the sentence question, accepts 92% of its 3,000-candidate sample (AUPRC 0.566 at a base rate of 0.476); asked about the pair, it scores 0.813, still behind the trained pair-conditioned model (0.860 on the same sample).

At 0.6B the three questions sit close to the base rate (0.66–0.68 against 0.574; 0.52 against 0.476): at that size the model barely does the task, what-

			AUPRC, sentences naming		EP
Model	AUPRC	F1	2 concepts	≥ 3	F1 (%)
Sentence-level	$0.738_{\pm.004}$	0.718	**0.909**	0.727	59.5
Pair-conditioned	$\mathbf{0.843}_{\pm.004}$	**0.817**	0.902	**0.847**	**69.6**
+ entity markers	$0.840_{\pm.003}$	0.815	0.907	0.844	69.0
Zero-shot 32B teacher, 3,000-candidate sample					
Sentence question	0.566	0.662	0.921	0.555	n/a
Pair question	0.813	0.766	0.958	0.799	n/a

Table 5: BioRED, BioCreative VIII test set: 16,606 sentence-scope candidates from 400 abstracts, 0.463 positive, 538 from sentences naming two concepts. Headline rows repeat Table 1. AUPRC is the per-seed mean $\pm$ standard deviation over three seeds (columns by concept count: three-checkpoint ensembles); F1 is at the pre-specified threshold (mean of the checkpoints' development thresholds), the teacher's at its greedy answer. EP F1 is BioRED's document-level entity-pair F1 (mean over three seeds), a pair scoring the maximum over the sentences that co-mention it. Bold: best among the trained models. On the teacher's sample the trained models' three-checkpoint ensembles score 0.757 and 0.860. Pair-conditioned against sentence-level: 2,642 items fixed, 1,064 broken (McNemar $p = 5.9\times10^{-148}$).

	TaxaPairs, 437				BioRED, 3,000		
Qwen3	sent.	pair	triple	Δ	sent.	pair	Δ
0.6B	0.676	0.658	0.663	-0.018	0.522	0.523	$+0.000$
1.7B	0.769	0.808	0.844	$+0.039$	0.533	0.648	$+0.115$
4B	0.761	0.885	0.921	$+0.124$	0.546	0.745	$+0.199$
8B	0.819	0.909	0.947	$+0.090$	0.524	0.800	$+0.275$
14B	0.818	0.929	0.959	$+0.112$	0.555	0.805	$+0.250$
30B-A3B	0.797	0.927	0.956	$+0.130$	0.571	0.792	$+0.221$
32B (teacher)	0.804	0.960	0.964	$+0.156$	0.566	0.813	$+0.246$
Qwen3.5-122B-A10B	0.788	0.985	0.984	$+0.197$	0.519	0.820	$+0.301$
trained, same rows	0.871	0.947	0.948	$+0.076$	0.757	0.860	$+0.103$

Table 6: AUPRC of zero-shot Qwen3 (Qwen Team, 2025) by size, asked the three questions of Appendix A (4-bit builds served locally, thinking disabled; the triple question in both argument orders, taking the higher score). 30B-A3B is a mixture of experts with 3B active parameters; Qwen3.5-122B-A10B (Qwen Team, 2026), a later release, is a mixture of experts with 8 of 256 routed experts and one shared expert active per token, run partly on the CPU. Base rates 0.574 and 0.476. Last row: the trained three-checkpoint ensembles on the same rows. Δ: pair minus sentence question (last row: pair-conditioned minus sentence-level model).

ever it is asked. From 1.7B on, the pair question is ahead at each size on both benchmarks: by $+0.039$ AUPRC at 1.7B and by $+0.09$ to $+0.20$ from 4B on ($+0.11$, then $+0.20$ to $+0.30$, on BioRED). The sentence question gains little with size (0.76–0.82 on TaxaPairs from 4B on; 0.52–0.57 on BioRED throughout, near its base rate), so scale mostly improves the answers to the pair question. On TaxaPairs, asked about the pair without the passage, the Qwen3 models from 1.7B to 32B improve with size as much (Appendix Q), so on that benchmark this growth need not reflect better reading of the passage. The lead is not monotonic in size among the Qwen3 models, and it is largest for the 122B model: $+0.197$ on TaxaPairs and $+0.301$ on BioRED, where the 32B model's is $+0.246$. Asked the triple question, every Qwen3 model from 1.7B to 30B-A3B scores above its answer to the pair question on TaxaPairs, while the 0.6B, 32B and 122B models score level with it. The triple question also checks the relation's type and direction and is asked in both argument orders, so this lead need not come from the relation term alone. Asked about the pair, the 122B model ranks the 437 candidates better than any trained model (0.985; the recommended model 0.970, the pair-conditioned student 0.942 per-seed mean, 0.947 as an ensemble), while soft labels and larger encoders raise the students little (Appendices F and H); asked the sentence question, the 122B model scores 0.788.

F Soft-label distillation

The students learn from the teacher's verdicts. We asked the teacher again for $P(\text{YES})$ on all 48,338 training rows (same prompt and inputs; its verdicts agree with the corpus labels on 99.7% of rows, and 24% of the probabilities lie between 0.05 and 0.95) and trained the three models again against these soft targets with a soft cross-entropy, selecting checkpoints on the same hard-labelled development split. It is the only use of the teacher's probabilities we tried, scored once (Table 7). Every model gains a little AUPRC ($+0.005$ to $+0.007$), the triple-conditioned model on every seed, and it also gains 0.021 F1 ($p = 0.031$ before correcting for three comparisons); against the pair-conditioned model it is marginally ahead (0.951 against 0.947; 25 items fixed, 21 broken, $p = 0.66$). The comparison is unchanged: the pair input lifts AUPRC from 0.873 to 0.947 (72 fixed, 28 broken, $p = 1.7 \times 10^{-5}$), by $+0.095$ on passages naming three or more taxa and $+0.036$ on the rest. Probabilities close about a quarter of the pair-conditioned model's distance to its teacher's 0.964 (0.016 AUPRC, against 0.021 with verdicts). The body keeps the verdict-trained models, on which every other analysis was run.

Model	Target	AUPRC	F1	AUPRC, taxa ≤ 2	AUPRC, taxa ≥ 3
Sentence	verdict	$0.867_{\pm.006}$	0.790	0.896	0.857
	$P(\text{YES})$	$0.873_{\pm.007}$	0.799	0.908	0.862
Pair	verdict	$0.942_{\pm.007}$	0.882	0.939	0.955
	$P(\text{YES})$	$0.947_{\pm.003}$	0.883	0.944	0.957
Triple	verdict	$0.945_{\pm.005}$	0.867	0.930	0.963
	$P(\text{YES})$	$0.951_{\pm.003}$	0.888	0.936	0.965

Table 7: The sentence-level, pair-conditioned and triple-conditioned models trained on the teacher's verdicts (the body's models) and on its probabilities, recipe and seeds unchanged. AUPRC is the per-seed mean $\pm$ standard deviation; F1 (block-held-out thresholds) and AUPRC by the number of taxa the passage names are those of three-checkpoint ensembles, as in Tables 1 and 4. Per-seed AUPRC with soft targets: .866/.879/.875, .951/.946/.945 and .952/.952/.948. Against the verdict-trained model of the same format, the soft-target model fixes 13 items and breaks 7 (sentence-level), 13 and 14 (pair-conditioned), 19 and 7 (triple-conditioned; McNemar $p = 0.26$, 1.0, 0.031).

Segment A given to the model	P	R	F1	AUPRC
Retrieved triple	0.842	0.892	**0.867**	**0.948**
Generic relation, pair kept	0.918	0.673	0.777	0.906
Relation shuffled across items	0.900	0.721	0.801	0.930
Relation only, no taxa	0.833	0.020	0.039	0.691
Empty segment A	1.000	0.036	0.069	0.649

Table 8: Degrading segment A of the trained triple-conditioned model at inference, weights and its own block-held-out thresholds unchanged; AUPRC of the three-checkpoint ensemble. Base rate 0.574. Per-seed standard deviations on AUPRC are ≤ 0.032. Bold: best F1 and AUPRC.

G Input ablation

Degrading the trained triple-conditioned model's segment A at inference, with weights and thresholds unchanged, shows which part of segment A it uses (Table 8; §4.5). Deleting both taxa collapses ranking from 0.948 to 0.691 against a base rate of 0.574, and emptying segment A entirely gives 0.649; replacing or shuffling the relation, pair kept, moves ranking far less (0.906, 0.930): the model relies mainly on the pair, which is what the pair-conditioned model already showed.

The empty-segment variant is not the sentence-level model: deprived of the input it was trained with, it does far worse (0.649) than a model trained without one (0.871; both are three-checkpoint ensembles), which is why each formulation has its own trained model.

Encoder	BioRED, BC8 test sent.	pair	EP F1 (%)	TaxaPairs, 437 sent.	pair	triple	pair F1
BiomedBERT-base	0.738	0.843	**59.5**/69.6	0.867	0.942	0.945	0.882
BiomedBERT-large	0.741	0.843	59.0/69.7	n/a	n/a	n/a	n/a
BioLinkBERT-base	**0.743**	**0.855**	58.3/70.2	**0.888**	0.939	0.949	0.860
BioLinkBERT-large	0.737	0.848	58.0/**70.6**	0.881	**0.949**	**0.951**	**0.896**

Table 9: The three formulations under the encoders that were scored on test, recipe otherwise that of §3.2: AUPRC and EP F1 (sentence-level/pair-conditioned) as per-seed means over three seeds, and block-held-out F1 of three-checkpoint ensembles. BiomedBERT-large was trained on our corpus only for the seed-1 development screen and was not scored on TaxaPairs (Appendix P). Against the base encoder's model of the same format, BioLinkBERT-base's pair-conditioned model fixes 621 BioRED items and breaks 537 (McNemar $p = 0.015$) and on TaxaPairs 19 and 32 ($p = 0.093$); its sentence-level model on TaxaPairs, 42 and 20 ($p = 0.0077$). No BioLinkBERT-large model differs significantly from the base encoder's on TaxaPairs ($p \geq 0.23$).

H Encoders

Besides the recipe of §3.2 we screened seven others on development data: five epochs at learning rates 1, 2 and 3 $\times 10^{-5}$, BiomedBERT-large at 1 and 2 $\times 10^{-5}$, and BioLinkBERT-base and -large, pretrained on PubMed with citation links (Yasunaga et al., 2022). The three that beat the base recipe by 0.01 development AUPRC or more on both splits (seed 1, pair-conditioned model) were scored once on the test sets: BiomedBERT-large at 2×10^{-5} on BioRED only (Appendix P), and BioLinkBERT in its base size under the base recipe and in its large size (below). BioLinkBERT-base's development AUPRC is 0.910 for the BioRED pair-conditioned model over three seeds (base encoder 0.892, BiomedBERT-large 0.905), and 0.852 for the pair-conditioned model on our corpus on seed 1, then 0.843 and 0.833 on seeds 2 and 3. Its BioRED pair-conditioned model gains on every seed (AUPRC 0.853–0.858 against 0.839–0.847; EP F1 70.1–70.4 against 69.2–70.0), the best sentence-scope AUPRC we have on BioRED (Table 9). On TaxaPairs its pair-conditioned model does not gain (0.939 ± 0.004 against 0.942 ± 0.007; F1 0.860 against 0.882) while its sentence-level model does (0.888, F1 0.816 against 0.790), so the formulation gap narrows to +0.051 AUPRC (53 items fixed, 37 broken, $p = 0.11$), still concentrated on passages naming three or more taxa (+0.073 against +0.009). On BioRED it widens, to +0.113.

BioLinkBERT-large, at 2×10^{-5} for three epochs

like BiomedBERT-large, passed the same screen (development AUPRC 0.907 on BioRED and 0.869 on our corpus's development split, seed 1; three-seed means 0.903 and 0.862). Its pair-conditioned models gain a little on both benchmarks: on BioRED, AUPRC 0.848 ± 0.009 and EP F1 70.6 ± 0.7 (seeds 70.2–71.4, each above every base seed); on TaxaPairs, 0.949 ± 0.003 and F1 0.896. Neither gain is significant: against the base encoder's pair-conditioned model it fixes 652 BioRED items and breaks 653 ($p = 1.0$), and on TaxaPairs 28 and 22 ($p = 0.48$). Its other models, trained afterwards and scored once, complete the comparison: the sentence-level models reach 0.881 on TaxaPairs and 0.737 on BioRED (EP F1 58.0), the triple-conditioned model 0.951, and the pair input again wins, by +0.068 on TaxaPairs (74 items fixed, 28 broken, $p = 8.4 \times 10^{-6}$; +0.090 on passages naming three or more taxa, +0.026 on the rest) and by +0.111 on BioRED. We keep the base encoder: BioLinkBERT-base does not help the pair-conditioned model on TaxaPairs, and BioLinkBERT-large's gains are not significant on either benchmark.

I Recall on rejected candidates

REJECT50 asks whether candidates an earlier filter rejected contain recoverable positives, and it is too small to rank the models: 11 positives, so one item moves recall by nine points. At block-held-out thresholds the sentence-level model recovers 9 of the 11 by re-admitting 13 of the 30 correct rejections (F1 0.545), the pair-conditioned model recovers 6 while re-admitting 5 (0.545), and the triple-conditioned model recovers 8 while re-admitting 3 (0.727). We read only the qualitative point: the discarded candidates do contain recoverable interactions, and a model that reads the candidate finds them with fewer wrong re-admissions than one that reads only the passage. Most of these candidates fail before any verifier runs: 27 of the 41 carry an incorrect interaction term and 12 a mis-resolved taxon, 10 of them both.

J Error examples

The examples illustrate the mechanisms of §4.6 (Table 10).

K Entity disambiguation and direction

Taxonomy gate. The natural remedy for entity errors is to gate candidates on taxonomic validity. Against the Open Tree Taxonomy (Rees and Cranston, 2017) such a gate catches 1 of the 32 unsupported BIOTX100 candidates and discards 5 of the 65 supported ones. This is because *Illiger*, Dikarya, *copia* and *sulla*, the surnames, clades, transposon families and vernacular names behind the entity failures, all resolve in the taxonomy. The failure is therefore one of disambiguation, of the kind species-name normalisers address (Gerner et al., 2010), which a validity gate cannot detect; and the gate's casualties are viruses, which are legitimate partners in this domain. The *authority* and *same_organism* rules (Appendix L) target the disambiguation directly and handle these cases better.

Direction labels. The benchmark's expert grading marks reversed triples as correct; we verified two by hand, (*Breinlia*, host of, *Rattus*) and (*Demidospermus*, parasitized by, *A. inermis*). No directional number on the benchmark would therefore measure the model rather than the convention, which is why direction is scored on its own annotation (Appendix M). The supervision itself is also uneven: among the 5,591 reversed training rows, the rate at which the teacher rejects a reversal is governed by the relation's surface string. For the relation strings with at least 40 reversed rows, the rejection rate runs from 0.04 to 0.70 (*affecting* 0.70, *affected by* 0.61, *pollinators* 0.04), and strings of the same form can differ widely (*parasite of* 0.57, *host of* 0.08). Whether a training label encodes direction therefore depends on which string retrieval matched.

L Candidate rules

Several error mechanisms of §4.6 concern the candidate rather than the passage (whether two strings name one organism, whether a word is a surname, whether *relative to* is an interaction), and a classifier handles them poorly. We address them with eight deterministic rejection rules, applied to the candidate before the verifier decides, each targeting one of those mechanisms. The rules *same_organism* (apposition of a common and a scientific name, or identical forms), *own_clade* (an adjacent clade annotation such as "(Acari: Prostigmata)"), and *authority* (a taxonomic author parsed as a taxon) detect the same-organism and entity classes. *non_biotic_relation* rejects phylogenetic, comparative and biogeographic relation terms; *negated* rejects the two negation forms whose scope provably contains one

Mechanism	Candidate triple	Passage (abridged)
Pair-binding	(*T. gondii*, parasite of, swine)	"The entero-pathogenic *Cystoisospora suis*, a coccidian **parasite of swine** and close relative to *Toxoplasma gondii* …" (PMID 34193323)
Co-listing	(katydids, prey, beetles)	"… remains were from insects … (e.g. **beetles**, dragonflies, cockroaches, and female **katydids**), indicating that eavesdropping on prey signals is not the only prey-finding strategy used by this *bat*." (PMID 28261664)
Negation	(wheat, hosts, *F. acuminatum*)	"… *F. acuminatum* in New Zealand was found only on **non-wheat hosts**." (PMID 20943139)
Same organism	(*P. chinensis*, pollinators, Chinese knotweed)	"… potential pollinators of **Chinese knotweed (*Persicaria chinensis*)**." (PMID 25913190)
Entity	(**Siberia**, migrating, birds)	"… a portion of birds **migrating** from **Siberia** have common winter nesting areas." (PMID 17087198)

Table 10: One example of each mechanism among the pair-conditioned model's false positives; the entity example also shows a relation term that names no biotic interaction (*migrating*). Bold marks the words behind each error; scientific names are italic, as is the relation's true argument where it is not a candidate taxon (*bat*); in the entity case a place name was matched as a taxon.

System	P	R	F1	fixed	broken
Pair-conditioned	0.869	**0.896**	0.882	n/a	n/a
+ 7 rules	0.878	0.888	**0.883**	3	2
+ 8 rules	**0.887**	0.876	0.882	6	5
Triple-conditioned	0.842	**0.892**	0.867	n/a	n/a
+ 7 rules	0.860	0.880	0.870	6	3
+ 8 rules	**0.883**	0.869	**0.876**	13	6

Table 11: Deterministic candidate rules applied before the model's decision ("+ 7 rules": all but *co_listed*), all 437 rows, three-checkpoint ensembles at block-held-out thresholds. "Fixed" and "broken" are relative to the same model without rules. McNemar: $p = 1.0$ and 1.0 on the pair-conditioned model, 0.50 and 0.17 on the triple-conditioned model.

of the pair ("non-s hosts", "no infections with … s"); *not_an_organism* rejects organs and syndromes; *pathogen_modifier* rejects an adjective that only ever occurs inside a pathogen's name; and *co_listed* rejects two taxa that are conjuncts of one dependency-parsed list not governed by an interaction noun ("interactions between A and B" is kept; parser: spaCy `en_core_web_sm`, Honnibal et al., 2020).

Table 11 gives their effect on both models. On the pair-conditioned model they raise precision by 0.009 to 0.018 and lower recall by 0.008 to 0.020, F1 unchanged ($p = 1.0$): of its 34 false positives they catch 6, because a model told which pair it is judging already rejects most of what they target. On the triple-conditioned model they do more (precision $0.842 \rightarrow 0.883$, $p = 0.17$). Both are upper estimates, since every rule was written with errors on this benchmark in view (below); their main practical benefit is transparency, because each rejection names its rule.

Validation. No rule has a parameter fitted to evaluation labels, but rules can still be fitted by choice. Four of these (*non_biotic_relation*, *negated*, *not_an_organism* and *pathogen_modifier*) were written after reading the verifiers' false positives on this benchmark (§4.6), *co_listed* was added after the other seven had been evaluated on it, and *same_organism*, *own_clade* and *authority* were written earlier, from an earlier filter's errors on BIOTX100 and REJECT50. To keep their selection from simply memorising the benchmark, every rule had to pass a criterion fixed before it was evaluated: on the 42,747 training rows in retrieved argument order, at least 85% of the candidates it rejects must be teacher-negative, matching the teacher's own accuracy. Two first drafts failed it and were narrowed on training data alone (a general negation cue fired on "without affecting the host", and a role-noun list rejected "viruses" and "fungi", which are legitimate coarse taxa), and, of the candidates each final rule rejects, 0.88 to 1.00 are teacher-negative (*co_listed*, which needs a dependency parse, on a fixed 6,000-row sample). The first seven rules were then frozen, recorded by hash, and evaluated once; *co_listed* was written after that evaluation, then frozen and evaluated in the same way. On the 100 fresh candidates of §4.7, the four rules written after reading the false positives reject one candidate (a negative) and *co_listed* rejects six (five negatives), too few to confirm or contradict their effects.

Expert-graded label	n	directed	correct
forward or reverse	84	62 (74%)	55 of 62 (0.887)
undecidable (“?”)	12	n/a	9 left UNCERTAIN
bidirectional	1	n/a	1 BIDIRECTIONAL

Table 12: The recommended model's four-way direction output against 97 expert-graded direction labels. “Directed” counts FORWARD or REVERSE; the rest are UNCERTAIN. The abstention level (0.60) was chosen by one of the authors after seeing this set's coverage curve, so the figures are optimistic.

M The recommended configuration

The model we recommend for use (§5) lies outside the controlled comparison. It is one 110M-parameter model with two heads, and it differs from the pair-conditioned model of §3.2 in three respects adopted during development: it gives the two taxa in alphabetical order, so the input is identical whichever way round retrieval presents the pair; it marks them in place in the passage (@…@, #…#) rather than only naming them in segment A; and its training labels are the teacher's pair-level judgements. Its interaction threshold, 0.5, was set before evaluation and never tuned. On the 437 candidates it reaches AUPRC 0.970 and F1 0.915 (precision 0.873, recall 0.960), and with the candidate rules of Appendix L it reaches precision 0.888, recall 0.948 and F1 0.917 (5 items fixed, 3 broken, $p = 0.72$).[2] It scores 32.3 candidates per second on eight CPU threads, measured with both heads. Escalating the 30% of candidates nearest its threshold to a zero-shot Qwen3.5-122B-A10B raises its precision with the rules from 0.888 to 0.962 at recall 0.912 (Appendix N).

Direction head. The second head reports the interaction's direction in four categories. FORWARD and REVERSE say which taxon acts on the other; they come from a direction score that is exactly antisymmetric in the two marked taxa, and are reported only when it is confident; otherwise the label is UNCERTAIN. BIDIRECTIONAL is assigned when the relation is mutual by definition (mutualism, symbiosis, competition, co-infection, *interacts with*), as listed in a relation-polarity lexicon derived from the Relation Ontology (Smith et al., 2005); for those, neither taxon is the subject and the model is not consulted. The benchmark's expert grading scores reversed triples as correct (Appendix K), so direction is evaluated on 97 separately expert-graded candidates (Table 12): the model assigns a direction to 74% of the directional ones, correctly on 0.887 of those, an optimistic figure since the abstention level was chosen on that set.

[2]The *co_listed* rule, added after the other seven were frozen, is omitted from this configuration: on this model's accept set it removes one false positive and three true ones.

		Qwen3-32B			Qwen3.5-122B-A10B		
Band	LLM calls	P	R	F1	P	R	F1
0%	0	0.888	0.948	0.917	0.888	0.948	0.917
10%	44	0.908	0.940	0.924	0.912	0.944	0.928
20%	87	0.920	0.920	0.920	0.932	0.924	0.928
30%	131	0.950	0.912	0.931	0.962	0.912	0.937
50%	218	0.964	0.849	0.903	0.981	0.841	0.906
100%	437	0.960	0.773	0.857	0.984	0.757	0.856

Table 13: The recommended model with its candidate rules keeps its own decision except on the band of candidates whose score lies nearest its threshold, which take a zero-shot LLM's answer to the pair question (Qwen3-32B, the teacher, or Qwen3.5-122B-A10B); 437 rows. A rule rejection is final. Band 0% is the recommended model alone, 100% the LLM alone (with the rules).

N Escalating uncertain candidates to a zero-shot LLM

The recommended model is fast and recall-oriented, and a large zero-shot LLM, asked about the pair, is precise but slow (Appendix E). Table 13 combines them: the model decides every candidate, and those it is least sure of (a fixed fraction, nearest its threshold) are re-decided by a zero-shot LLM. Escalating 30% to Qwen3.5-122B-A10B raises precision from 0.888 to 0.962 at recall 0.912, removing 21 of the 30 false positives; the model alone reaches that precision only at $\tau = 0.99$, where recall falls to 0.745. Against the model alone the decisions are not significantly better on these 437 rows (25 items fixed, 13 broken, McNemar $p = 0.074$; with Qwen3-32B, 22 and 13, $p = 0.18$): the gain lies in the operating point, with more recall at a given precision, while accuracy at the model's own threshold does not improve significantly. We report the band at six settings and do not choose one. Its cost is set by the LLM's speed: escalating 30% of the 175,588-candidate pool means about 53,000 calls, under a day on one A100 for Qwen3.5-122B-A10B (1.4 s per call with part of it on the CPU) and less for the 32B.

O Artefacts and compute

The training recipe of §3.2 is AdamW (Loshchilov and Hutter, 2019) at learning rate 2×10^{-5} and

weight decay 0.01, a linear schedule with 10% warm-up, batch size 32, inputs of at most 256 wordpieces (512 for the whole-abstract BioRED model of §4.2), three epochs and gradient clipping at 1.0; Appendix H lists the alternatives we screened. The encoders are BiomedBERT-base (`microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`) and -large (MIT licence) and BioLinkBERT-base and -large (Apache 2.0); the zero-shot models are Qwen3 0.6B–32B (Qwen Team, 2025) and Qwen3.5-122B-A10B (Qwen Team, 2026), all Apache 2.0, run locally through Ollama; taxon mentions are counted with TaxoNERD (MIT). BioRED and its BioREDirect directionality release are public NCBI resources, used as distributed. We will release our grades of TaxaPairs, the labels of the 100 fresh candidates and the teacher's labels of the training corpus under CC BY 4.0, and the code under the MIT licence. The candidate triples come from BiotXplorer (SIBiLS, CC BY 4.0). The passages are PubMed text (titles, abstracts and a few keyword lists), courtesy of the U.S. National Library of Medicine; their copyright remains with the publishers or authors, we distribute them with their PMIDs under NLM's terms, and a copy may differ from the current PubMed record. The teacher labels were produced with Qwen3-32B (Apache 2.0). Eleven training rows are conflict-of-interest statements, some naming people; the release leaves them out. Unless stated otherwise, everything ran on one NVIDIA A100 (80 GB): a three-epoch encoder run takes between one and twenty-five minutes (the two eight-epoch runs of Appendix P longer), and the whole study took about 50 GPU-hours, most of them LLM inference for labelling the corpus and for the zero-shot rows. The exceptions ran on CPUs: the three-epoch pair-only models of Appendix Q (training and scoring), the zero-shot answers without the passage reported in that appendix, and the BioRED scores behind the entity-pair F1 of §4.2; Qwen3.5-122B-A10B ran partly on the CPU.

P Negative results

Table 14 collects interventions that did not help, because each is a plausible first thing to try; the taxon-validity gate of its first row is described in Appendix K. Self-distillation at equal capacity is null, suggesting the teacher's knowledge is already saturated in the student. Compressing to six layers speeds up CPU inference, which this setting does not need, and lowers precision by 0.088. Reweighting the corpus towards the signal distribution of the retrieval pool lowered AUPRC by 0.059. The fifth row is the most informative. Before the candidate rules existed, we turned the same mechanisms into training data (synthetic negatives for the same-organism, own-clade and authority classes, and in a second variant for co-listing) and retrained. The first variant left AUPRC unchanged within seed noise and the second lowered it by 0.037. For properties of the candidate itself, rules worked better than extra training data. Deltas in rows two to five were measured under an earlier version of the evaluation and are given as directions. The last row is one of the three recipes of §4.5 that passed the development screen (the others are in Appendix H): BiomedBERT-large raises the BioRED pair-conditioned model's development AUPRC from 0.892 to 0.905 (three-seed means), but on the BC8 test set the pair-conditioned model scores 0.843 ± 0.005 against 0.843 ± 0.004, F1 0.814 against 0.817, EP F1 69.7 ± 0.5 against 69.6 ± 0.4, and the sentence-level model 0.741 against 0.738; the pair input's gain is unchanged ($+0.103$ against $+0.105$ AUPRC). Beyond the development screen it was not trained on our corpus or scored on TaxaPairs, since its BioRED test result could not change the choice. For the whole-abstract BioRED model of §4.2 we also tried, on devel-

Intervention	Benefit	Drawback
Taxon-validity gate	1 of 32 unsupported caught	$-5/65$ supported; rejects viruses
Self-distillation†	none	-0.001 precision, -0.008 F1
6-layer compression†	faster CPU inference	-0.088 precision
Corpus signal-matching†	none	-0.059 AUPRC
Rule-family training data†	none	AUPRC $+0.005$ to -0.037
BiomedBERT-large	$+0.013$ development AUPRC (BioRED pair)	none on test; $3\times$ parameters

Table 14: Six plausible design choices whose drawbacks outweighed their benefits in our measurements. †Measured under an earlier version of the evaluation; deltas are directions only (see text). The last row compares per-seed means.

opment data only (seed 1), eight epochs instead of three, alone and with each concept's BioRED type in segment A: development AUPRC 0.786 and 0.791 against 0.786 for three epochs on the same seed, gains of at most 0.005, smaller than the 0.021 spread of the three-epoch seeds (0.765–0.786), so neither was trained further or scored on the test set.

Q Pair-only controls

The pair-only model of §4.5 is the pair-conditioned model of §3.2 with the passage removed: it reads s_1 `[SEP]` s_2 as a single segment. Encoder, corpus, seeds, recipe, checkpoint selection and threshold policy are those of the pair-conditioned model on each benchmark; it is scored order-free like every trained model that reads the candidate (§3.5), and no model was scored more than once on a test set. Its per-seed AUPRC is 0.807, 0.798 and 0.819 on TaxaPairs and 0.714, 0.697 and 0.717 on BioRED. On TaxaPairs its development AUPRC was still rising at the third epoch, so we also trained it for up to ten epochs, recipe otherwise unchanged, keeping the best development epoch (the fourth, third and sixth for the three seeds). This longer-trained model reaches AUPRC 0.804 ± 0.011 (three-checkpoint ensemble 0.813, against 0.812 after three epochs), and the pair-conditioned model still leads it by 0.133 (bootstrap 95% interval $[0.082, 0.186]$). All other pair-only results are for the three-epoch model. The teacher's answers without the passage, below, give a second names-only reference. On TaxaPairs the three-epoch model's block-held-out precision is 0.671 at recall 0.952, and its three-checkpoint ensemble's AUPRC is 0.926 on BIOTX100, 0.800 on TEST299 and 0.433 on REJECT50 (the other models' are in Table 4). Against the pair-only model, the pair-conditioned model fixes 95 items and breaks 26 (McNemar $p = 6.3 \times 10^{-10}$), and the sentence-level model fixes 80 and breaks 62 ($p = 0.15$). On BioRED its development thresholds are low (0.10 to 0.16), so its pre-specified operating point favours recall (precision 0.617, recall 0.859); against it, the pair-conditioned model fixes 2,786 items and breaks 745 ($p = 2.7 \times 10^{-258}$), and the sentence-level model, whose F1 is the same, fixes 2,906 and breaks 2,443 ($p = 2.7 \times 10^{-10}$). Its entity-pair F1, 59.7 ± 0.9 (precision 0.532, recall 0.683), is level with the sentence-level model's. On BioRED the two names also reveal the candidate's entity types, a cue the sentence-level model cannot use because it is not told the pair, so part of the pair-only model's score there, and of the pair-conditioned model's lead over the sentence-level model, may come from the types.

Seen and unseen pairs. A TaxaPairs pair counts as seen when, lowercased and in either order, it is the pair of some row of the 48,338-row training corpus, which holds 28,815 distinct pairs over 18,439 taxa; a taxon counts as seen when it is an argument of some training row. An argument that lists alternative names counts as seen if any of them matches (without this rule, 32 candidates have a seen pair instead of 38). Pairs are counted against the whole corpus, development split included, so 3 to 6 of the 38 (depending on the seed) were seen only in development data. Matching is by surface string, as is the pair grouping of the development split (§3.2), so the same two taxa under other names (a synonym, an abbreviated genus) count as unseen here and can fall on both sides of that split; 38 is therefore a lower bound. Few benchmark pairs occur in training partly by design: the corpus's 20,000-row base sample was drawn after removing pool rows whose pair matched, lowercased, a pair of TEST299 or, through its canonical names, of BIOTX100; REJECT50 was not covered. On BioRED the training candidates of §3.4 define seen pairs, by normalised surface string or by concept identifier. Table 15 gives AUPRC on these subsets.

The 38 TaxaPairs candidates with a seen pair are too few to separate the pair-only and pair-conditioned models; on them, scoring each candidate by the share of its pair's training rows labelled positive, with no model, already reaches 0.912 (0.598 on all 437 rows, where unseen pairs take the corpus base rate). On the 399 candidates with an unseen pair the pair-conditioned model leads the sentence-level model by about as much as on all rows, and on the 98 candidates neither of whose taxa occurs in training it reaches 0.980, against 0.920 for the sentence-level model, while the pair-only model falls to 0.772. On BioRED, too, the lead over the sentence-level model holds on unseen pairs and the lead over the pair-only model widens where neither entity occurs in training; its 381 candidates with a seen pair again do not separate the pair-only and pair-conditioned models. Matching pairs by concept identifier instead (505 candidates with a seen pair) leaves the pair-conditioned model's lead on the other 16,101 candidates at $+0.106$ $[0.096, 0.116]$ over the sentence-

Candidates	n	Pos.	Sentence	Pair	Pair-only	Pair − pair-only	Pair − sentence
TaxaPairs							
all	437	0.574	0.871	0.947	0.812	$+0.134$ $[0.085, 0.185]$	$+0.076$ $[0.048, 0.107]$
pair seen in training	38	0.711	0.898	0.945	0.957	-0.012 $[-0.067, 0.034]$	$+0.047$ $[0.001, 0.110]$
pair unseen	399	0.561	0.870	0.951	0.808	$+0.143$ $[0.088, 0.199]$	$+0.081$ $[0.050, 0.117]$
neither taxon seen	98	0.582	0.920	0.980	0.772	$+0.209$ $[0.099, 0.318]$	$+0.061$ $[0.016, 0.120]$
BIOTX100, GloBI half	50	0.860	0.986	0.994	0.964	$+0.030$ $[0.003, 0.073]$	$+0.008$ $[-0.013, 0.042]$
BIOTX100, random half	47	0.702	0.920	0.959	0.876	$+0.083$ $[-0.004, 0.194]$	$+0.039$ $[0.005, 0.088]$
BioRED, BioCreative VIII test set; pairs matched by surface string							
all	16,606	0.463	0.742	0.851	0.725	$+0.126$ $[0.116, 0.137]$	$+0.109$ $[0.100, 0.119]$
pair seen in training	381	0.307	0.478	0.653	0.664	-0.011 $[-0.079, 0.058]$	$+0.175$ $[0.101, 0.246]$
pair unseen	16,225	0.466	0.748	0.854	0.726	$+0.128$ $[0.118, 0.138]$	$+0.106$ $[0.096, 0.116]$
neither entity seen	8,427	0.414	0.765	0.889	0.740	$+0.148$ $[0.134, 0.163]$	$+0.123$ $[0.110, 0.137]$
BioRED, pairs matched by concept identifier							
pair seen in training	505	0.372	0.591	0.744	0.714	$+0.030$ $[-0.023, 0.084]$	$+0.153$ $[0.093, 0.211]$
pair unseen	16,101	0.466	0.747	0.853	0.726	$+0.128$ $[0.117, 0.139]$	$+0.106$ $[0.096, 0.116]$

Table 15: AUPRC of the three-checkpoint ensembles on candidate subsets defined by the training data (TaxaPairs: the 48,338-row corpus; BioRED: its training candidates) and on the two halves of BIOTX100 (§3.3). Seen: the pair, or a taxon or entity, occurs in training, matched by surface string or, on BioRED, also by concept identifier. Pos.: share of positives, the AUPRC of a random ranking. The last two columns give AUPRC differences with paired bootstrap 95% intervals (2,000 resamples of candidates within the subset, fixed seed).

Qwen3	sentence	pair	no passage	Δ
1.7B	0.769	0.808	0.690	$+0.118$
4B	0.761	0.885	0.767	$+0.118$
8B	0.819	0.909	0.803	$+0.106$
14B	0.818	0.929	0.827	$+0.102$
30B-A3B	0.797	0.927	0.792	$+0.134$
32B (teacher)	0.804	0.960	0.871	$+0.089$
trained, same rows	0.871	0.947	0.812	$+0.134$

Table 16: AUPRC of zero-shot Qwen3 models from 1.7B to 32B on the 437 TaxaPairs candidates (base rate 0.574), asked the sentence and pair questions (as in Table 6) or the no-passage question. Δ: pair question minus no-passage question. Last row: the trained three-checkpoint ensembles, with the pair-only model under "no passage". The no-passage answers were computed on CPUs, the other two columns on a GPU (Appendix O).

level model and $+0.128$ $[0.117, 0.139]$ over the pair-only model.

The two halves of BIOTX100. One half of BIOTX100 was drawn from pairs recorded in GloBI and the other at random (§3.3). Of the GloBI half's 50 clean rows, 8 have a pair seen in training, against 14 of the 47 in the random half. On the GloBI half the pair-only model already reaches 0.964 at a base rate of 0.86, so that half says little about reading the passage; the pair-conditioned model leads the pair-only model by 0.030 there and by 0.083 on the random half, where the interval includes zero (Table 15).

Zero-shot without the passage. The no-passage question, below, rewrites the pair question of Appendix A without the passage: it asks whether both strings name organisms and whether the two are known to interact directly, and it lists the sentence question's examples of interactions. Like the pair question, it is asked in one argument order, and it differs from the pair question in wording as well as in input (see the Limitations on prompt wording). Table 16 gives its AUPRC on TaxaPairs for the Qwen3 models from 1.7B to 32B. At every size the pair question, which adds the passage, scores 0.089 to 0.134 higher. Without the passage the 32B teacher scores 0.871, above its answer to the sentence question (0.804), so the names alone tell it much about which of these pairs interact: it scores 0.991 on the GloBI half of BIOTX100 and 0.833 on the random half (0.996 and 0.951 with the passage), and 0.883 on the 399 candidates whose pair is unseen in training (0.967 with it). The models' answers without the passage improve with size as much as their answers to the pair question (0.690 at 1.7B to 0.871 at 32B, against 0.808 to 0.960), so the growth of the pair question with size on TaxaPairs (Appendix E) need not come from reading the passage better.

```
You verify candidate biotic-interaction
pairs extracted from scientific
literature.

Candidate pair:
  Entity 1: "<taxon 1>"
  Entity 2: "<taxon 2>"
```

```
Answer YES only if ALL of the following
hold:
 1. Both surface strings genuinely refer
    to organisms (not a gene, protein,
    chemical, author name, or place).
 2. These two organisms are known to
    interact directly (for example
    predation, parasitism, infection,
    herbivory, pollination, mutualism,
    symbiosis or disease transmission),
    not merely through an interaction
    each has with some third organism.

Reply with exactly one word: YES or NO.
```